Department of Computer Science and Engineering (CSE)
BRAC University

Practice Problem Set 6

CSE251 – Electronic Devices and Circuits

BJT Circuits

Logic Function Implementation, Modes of Operation, Method of Assumed States, Multistage Circuits, MOSFET-BJT Hybrid Problems

I_B, I_C, and I_E denote the base, collector, and emitter currents, respectively. Similarly, V_B, V_C, and V_E denote the corresponding voltages. "Sat" and "Active" refer to the saturation and active regions, respectively. k_n and k_n' denote MOSFET's transconductance and process transconductance parameters, respectively. V_{T_n} denote the threshold voltage of the MOSFETs. β denote the common emitter current gain of BJTs.

Course Description, COs, and Policies

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Midterm and Final Questions

For any error or issues, contact purbayan.das@bracu.ac.bd

Problem 1

- Give a switch-BJT implementation of the following logic functions. A, B, C, D, E and F are the Boolean inputs and f is the Boolean output. Here, "+" and "." represent the logical OR and AND operations, respectively.

I. $f = A.\bar{B}$

II. $f = \overline{A.B.C + D.E}$

III. $f = \overline{A + B.C + D}$ [Summer 2024]

IV. $f = A.B + \bar{A}.\bar{B}$ [Fall 2024]

V. $f = \overline{A.C} + \overline{B + C}$

VI. $f = (A.B + C).D$

VII. $f = A.B + B.C + C.A$

VIII. $f = (A + B).(C + D)$

IX. $f = (A + B).C + D.E$

X. $f = \bar{A}.B + \bar{B}.A + \bar{C}.A$ [Spring 2025]

XI. $f = (A.B + C).D.(E + F)$

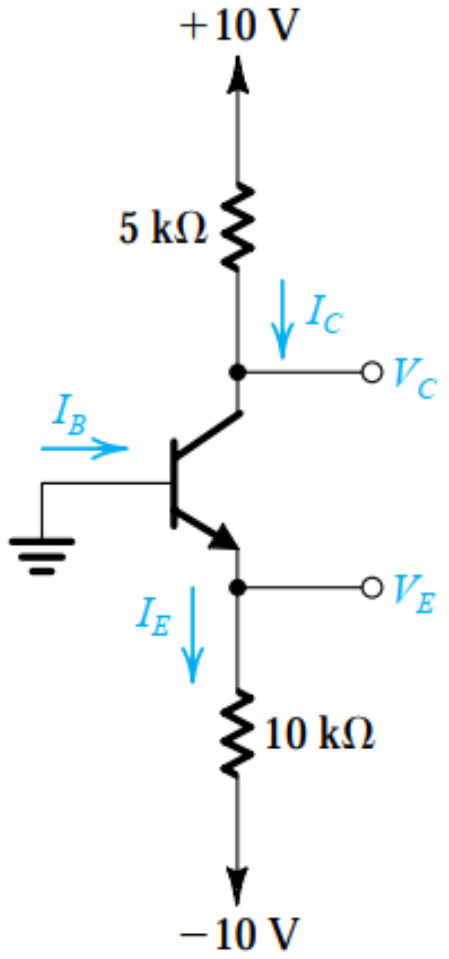
XII. $f = A \oplus B$

XIII. $f = \overline{(A + B + C).D + \bar{D}}$

XIV. $f = \overline{C.(A + B)}.(A + \bar{B} + C)$

Problem 2

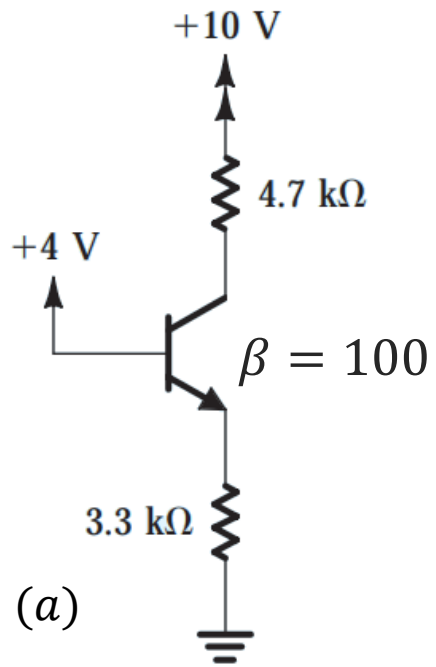
- For the adjacent circuit, V_E was measured and found to be $-0.7\ V$. If $\beta = 50$, determine I_E, I_B, I_C, and V_C.



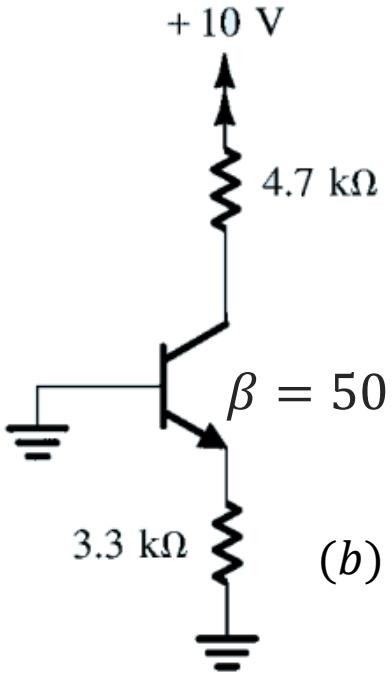
Ans: $\boldsymbol{Active; I_E = 0.93\ mA; I_B = 18\ \mu A; I_C = 0.91\ mA; V_C = 5.45\ V; V_E = -0.7\ V}$

Problem 3

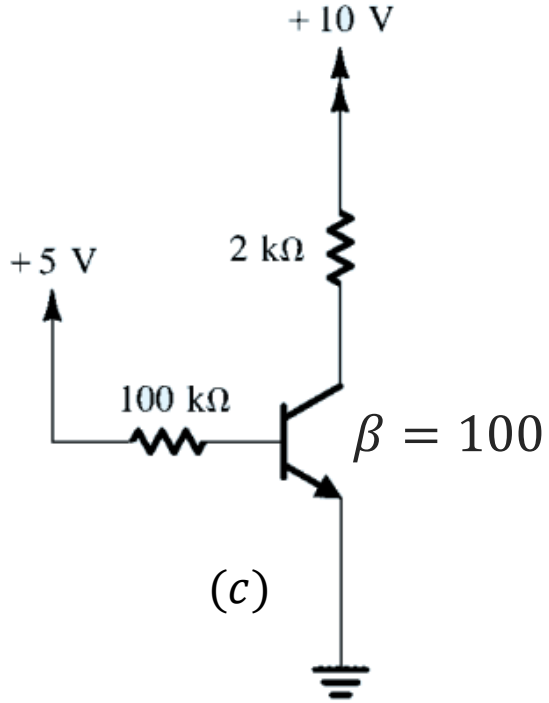
- Determine and verify the operating region for each of the transistors in the following circuits. Assume $V_{BE,\,active} = 0.7\,V$, $V_{BE,\,sat} = 0.8\,V$, and $V_{CE,\,sat} = 0.2\,V$.



Ans: $\boldsymbol{Active; I_C = 0.99\,mA; I_E = 1\,mA; I_B = 10\,\mu A; V_B = 0\,V;\ V_C = 5.347\,V; V_E = 3.3\,V}$



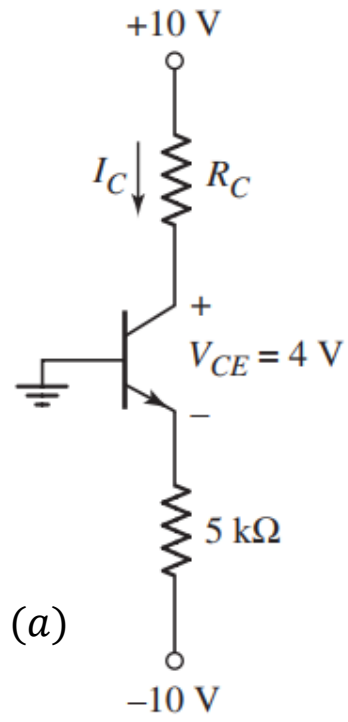
Ans: $\boldsymbol{Cutoff; I_C = I_E = I_B = 0; V_B = V_E = 0\,V;\ V_C = 10\,V}$



Ans: $\boldsymbol{Active; I_C = 4.3\,mA; I_E = 4.343\,mA; I_B = 43\,\mu A; V_B = 0.7\,V;\ V_C = 1.4\,V; V_E = 0\,V}$

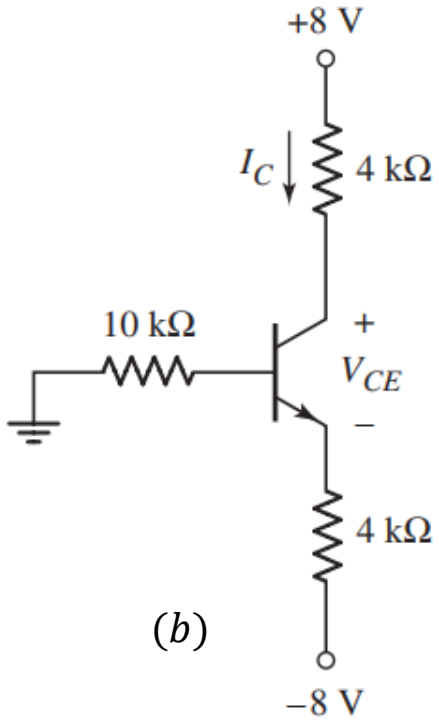
Problem 4

- For the NPN transistors in the following circuit, $\beta = 75$, $V_{BE,\,active} = 0.7\ V$, $V_{BE,\,sat} = 0.8\ V$, and $V_{CE,\,sat} = 0.2\ V$. Determine the labeled quantities.



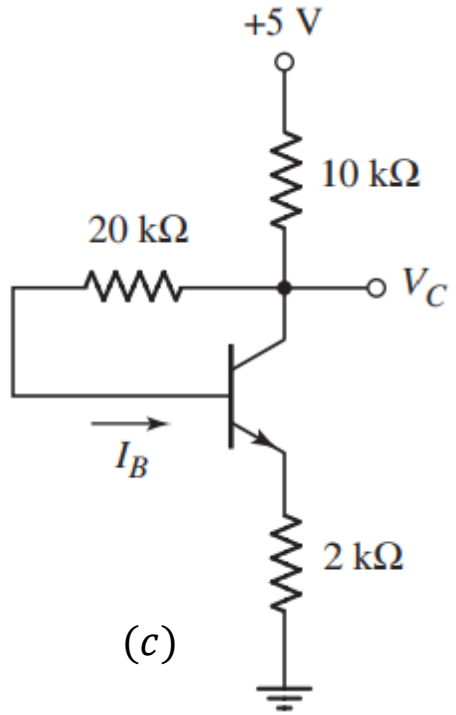
(a)

Ans: $\boldsymbol{Active; I_C = 1.84\ mA; I_E = 1.86\ mA; I_B = 20\ \mu A; V_B = 0\ V;\ V_C = 3.3\ V; V_E = -0.7\ V; R_C = 3.64\ k\Omega}$



(b)

Ans: $\boldsymbol{Active; I_C = 1.725\ mA; I_E = 1.76\ mA; I_B = 35\ \mu A; V_B = -0.26\ V;\ V_C = 1.1\ V; V_E = -0.96\ V}$

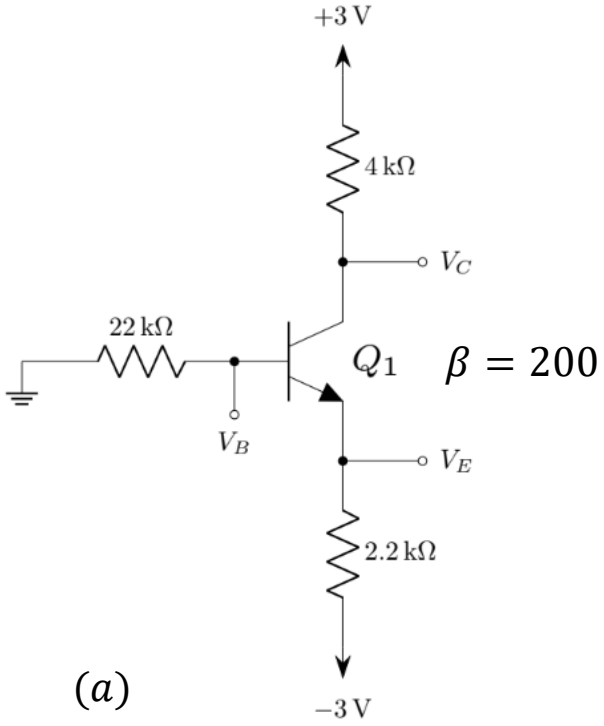


(c)

Ans: $\boldsymbol{Active; I_C = 0.341\ mA; I_E = 0.346\ mA; I_B = 5\ \mu A; V_B = 1.4\ V;\ V_C = 1.54\ V; V_E = 0.7\ V}$

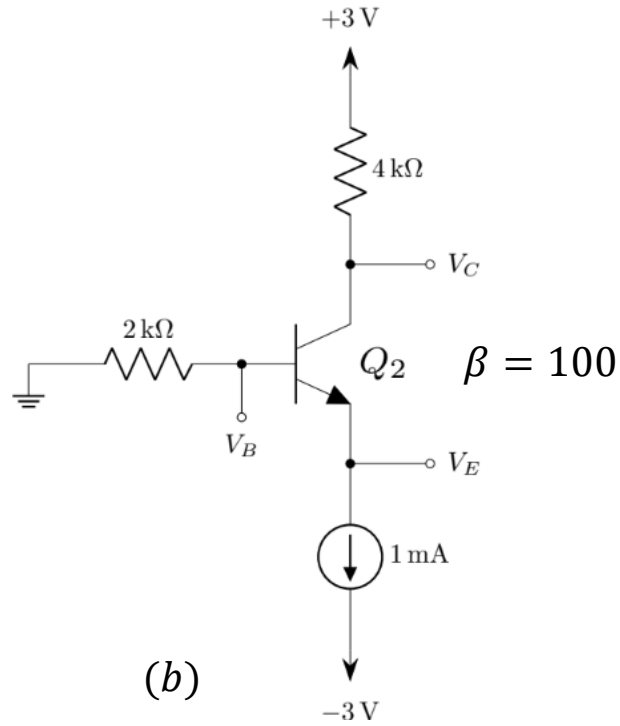
Problem 5

- For the transistors in the following circuits, $V_{BE,\,active} = 0.7\,V$, $V_{BE,\,sat} = 0.8\,V$, and $V_{CE,\,sat} = 0.2\,V$. Determine V_B, V_C, and V_E. *[Hint for (b): $V_{BC} = V_{BE} - V_{CE}$]*



(a)

Ans: $\boldsymbol{Sat; I_C = 0.933\ mA; I_E = 0.9394\ mA; I_B = 6.1\ \mu A; V_B = -0.132\ V;\ V_C = -0.732\ V; V_E = -0.932\ V}$

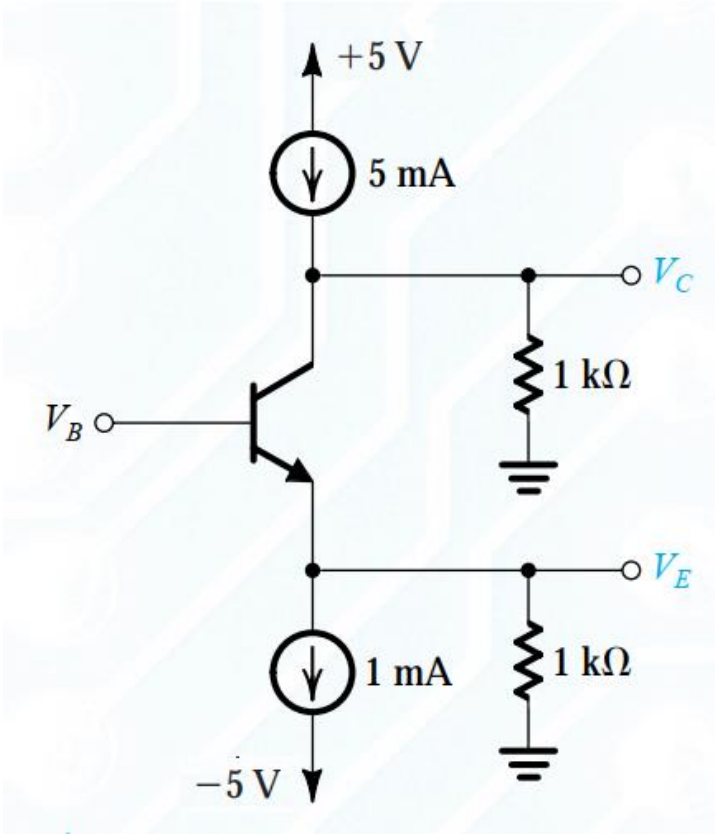


(b)

Ans: $\boldsymbol{Sat; I_C = 0.933\ mA; I_E = 1\ mA; I_B = 0.067\ mA; V_B = -0.134\ V;\ V_C = -0.732\ V; V_E = -0.934\ V}$

Problem 6

- For the adjacent circuit, assume that $V_{BE} = 0.7\,V$. For what value of V_B does the transistor operates in cut-off?



Ans: $\boldsymbol{I_C = I_B = I_E = 0;\ V_B = 0.3\,V;\ V_C = 5\,V; V_E = -1\,V}$

Problem 7

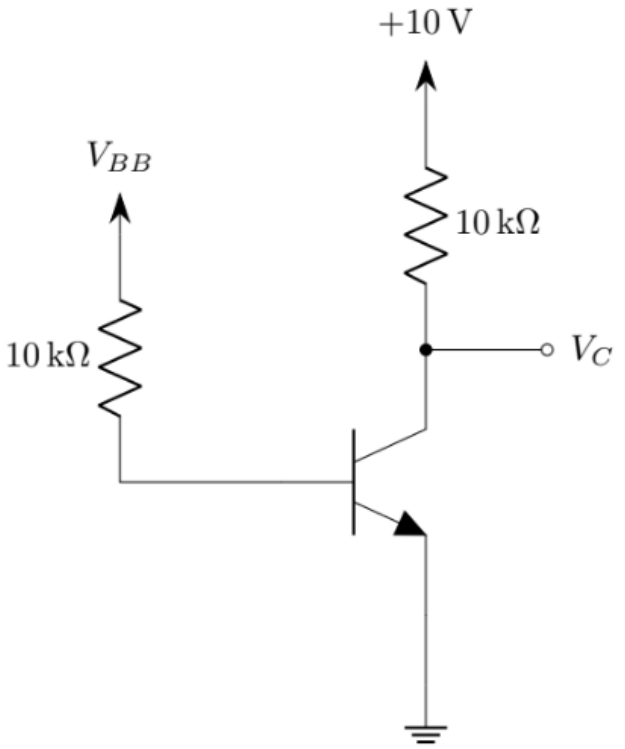
- For the adjacent circuit, assume that V_{BE} remains constant at $0.7\,V$ when the transistor is ON. The transistor β is specified to be 50. Determine the value of V_{BB} that results in the transistor operating:

  I. in the active region with $V_{CE} = 5\,V$,

  II. at the edge of saturation, and

  [Hint: at the edge of saturation, β is still 50 and assume $V_{CE} = 0.3\,V$.]

  III. deep in saturation with $\beta_{forced} = 10$ and $V_{CE} = 0.2\,V$.



Ans: $I.\ I_C = 0.5\,mA; I_E = 0.51\,mA; I_B = 10\,\mu A; V_B = 0.7\,V;\ V_C = 5\,V; V_E = 0\,V; V_{BB} = 0.8\,V$	Ans: $II.\ I_C = 0.97\,mA; I_E = 0.9894\,mA; I_B = 19.4\,\mu A; V_B = 0.7\,V;\ V_C = 0.3\,V; V_E = 0\,V; V_{BB} = 0.894\,V$	Ans: $III.\ I_C = 0.98\,mA; I_E = 1.076\,mA; I_B = 98\,\mu A; V_B = 0.7\,V;\ V_C = 0.2\,V; V_E = 0\,V; V_{BB} = 1.68\,V$

Problem 8

- For the following circuit, $V_S = 5\,V$, $R_L = 10\,k\Omega$, and $R_I = 500\,k\Omega$. Assume $\beta = 100$ for the transistor, $V_{BE,\,active} = V_{BE,\,sat} = 0.6\,V$, and $V_{CE,\,at\ the\ edge\ of\ sat} = V_{CE,\,sat} = 0.2\,V$.

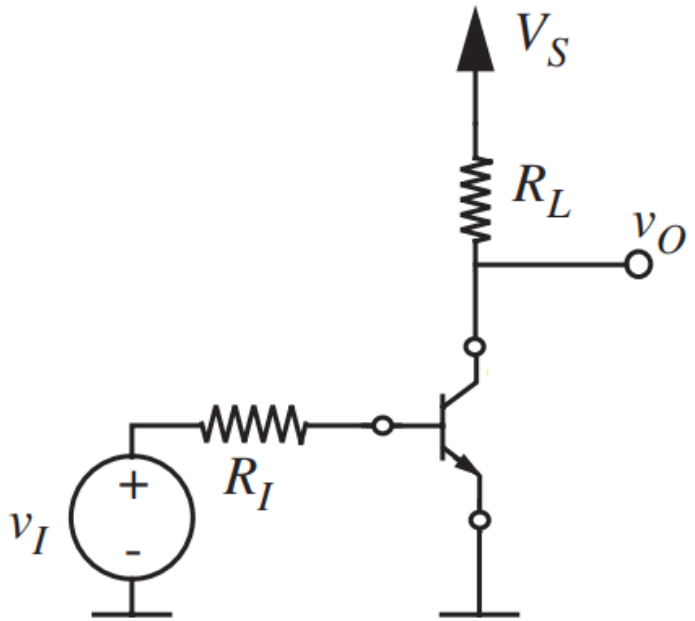
I. Derive an expression relating v_O and v_I assuming the transistor is in active mode. Ans: $\boldsymbol{v_0 = 6.2 - 2v_I}$

II. What is the highest value of v_I for which the transistor operates in cut-off. Ans: $\boldsymbol{0.6\,V}$

III. What is the highest (or lowest) value of v_I for which the transistor operates in its active region (or enters into its saturation region). What are the corresponding range for v_O? *[$\beta = 100$ at the edge of saturation]* Ans: $\boldsymbol{3\,V}$

IV. Based on the relation found in I and values of v_I and v_O found in II and III, draw the VTC ($v_O\ vs.\ v_I$) graph for the circuit.

Ans: $$v_0 = \begin{cases} 5\,V, & v_I \le 0.6\,V, \quad \textbf{Cutoff} \\ 6.2 - 2v_I, & 0.6\,V < v_I \le 3\,V, \quad \textbf{Active} \\ 0.2\,V, & v_I > 3\,V, \quad \textbf{Saturation} \end{cases}$$



Problem 9

- For the following circuit, $V_S = 10\ V$, $R_E = 100\ k\Omega$, and $R_I = 10\ k\Omega$. Assume $\beta = 100$ for the transistor, $V_{BE,\,active} = V_{BE,\,sat} = 0.6\ V$, and $V_{CE,\,at\ edge\ of\ sat} = V_{CE,\,sat} = 0.2\ V$.

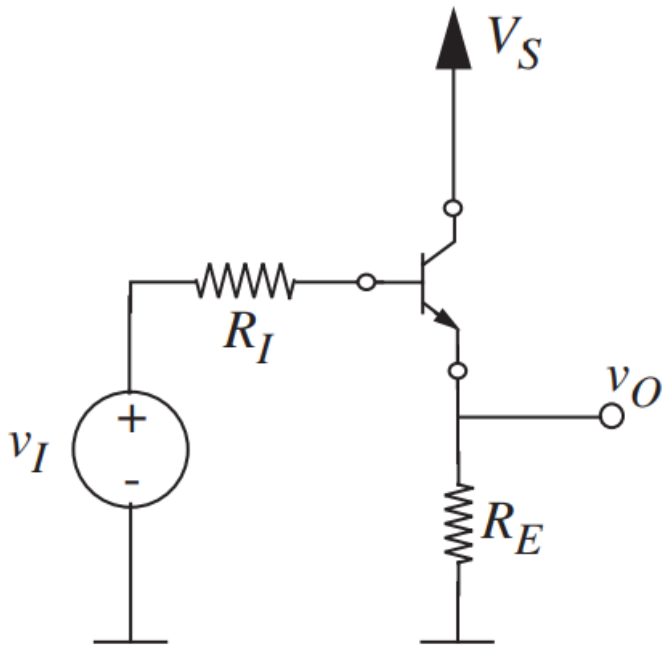
I. Derive an expression relating v_O and v_I assuming the transistor is in active mode. Ans: $\boldsymbol{v_0 \approx v_I - 0.6}$

II. What is the highest value of v_I for which the transistor operates in cut-off. Ans: $\boldsymbol{0.6\ V}$

III. What are the highest (or lowest) values of v_I for which the transistor remains in its active region (or operates at the edge of saturation). What are the corresponding range for v_O? *[$\beta = 100$ at the edge of saturation]* Ans: $\boldsymbol{10.4\ V}$

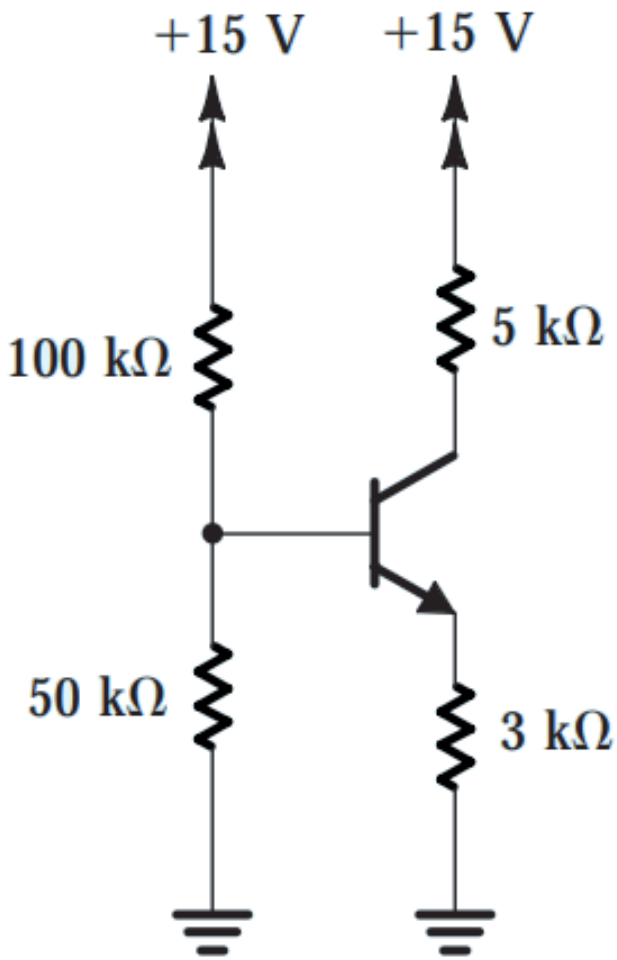
IV. Based on the expression found in I and values of v_I and v_O found in II and III, draw the VTC ($v_O\ vs.\ v_I$) graph for the circuit.

Ans: $$v_0 = \begin{cases} 0\ V, & v_I \leq 0.6\ V, \quad Cutoff \\ v_I - 0.6, & 0.6\ V < v_I \leq 10.4\ V, \quad Active \\ 9.8\ V, & v_I > 10.4\ V, \quad Saturation \end{cases}$$



Problem 10

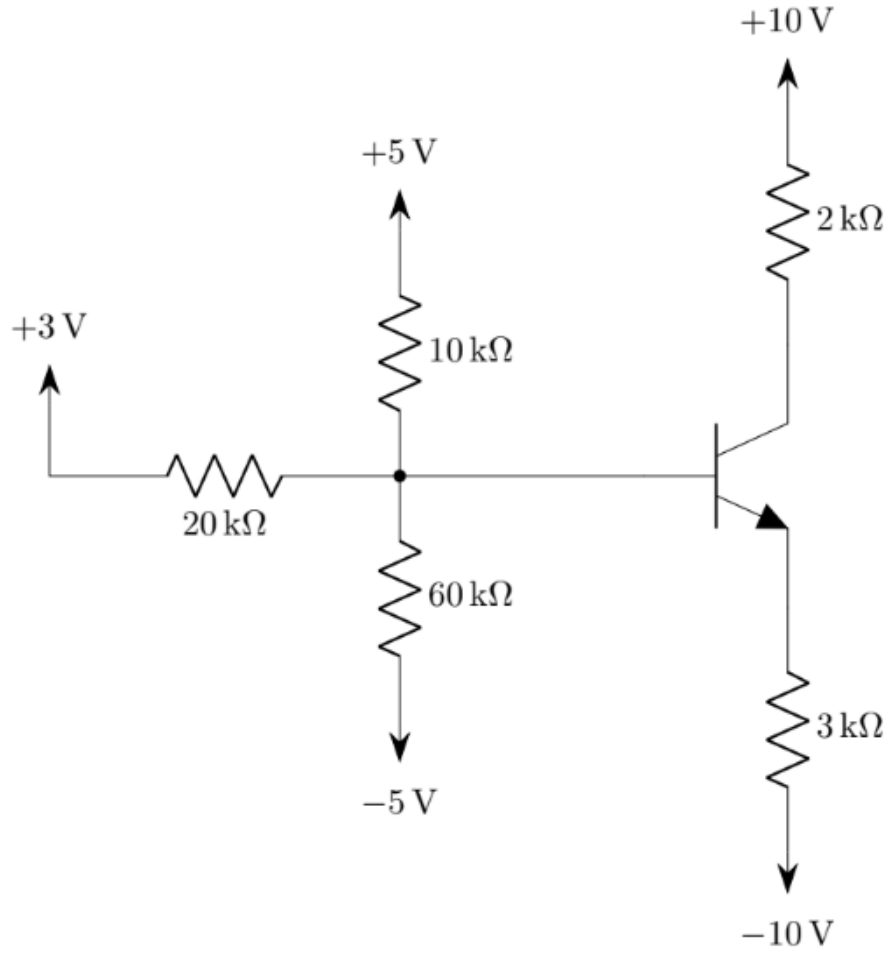
- For the NPN transistor in the following circuit, $\beta = 100$, $V_{BE,\,active} = 0.7\,V$, $V_{BE,\,sat} = 0.8\,V$, and $V_{CE,\,sat} = 0.2\,V$. Determine the current through the $50\ k\Omega$ resistor.



Ans: $\boldsymbol{Active;\ I_C = 1.28\,mA; I_E = 1.2928\,mA; I_B = 12.8\,\mu A; V_B = 4.58\,V;\ V_C = 8.6\,V; V_E = 3.88\,V;\ Current\ through\ 50\ k\Omega = 91.6\,\mu A}$

Problem 11

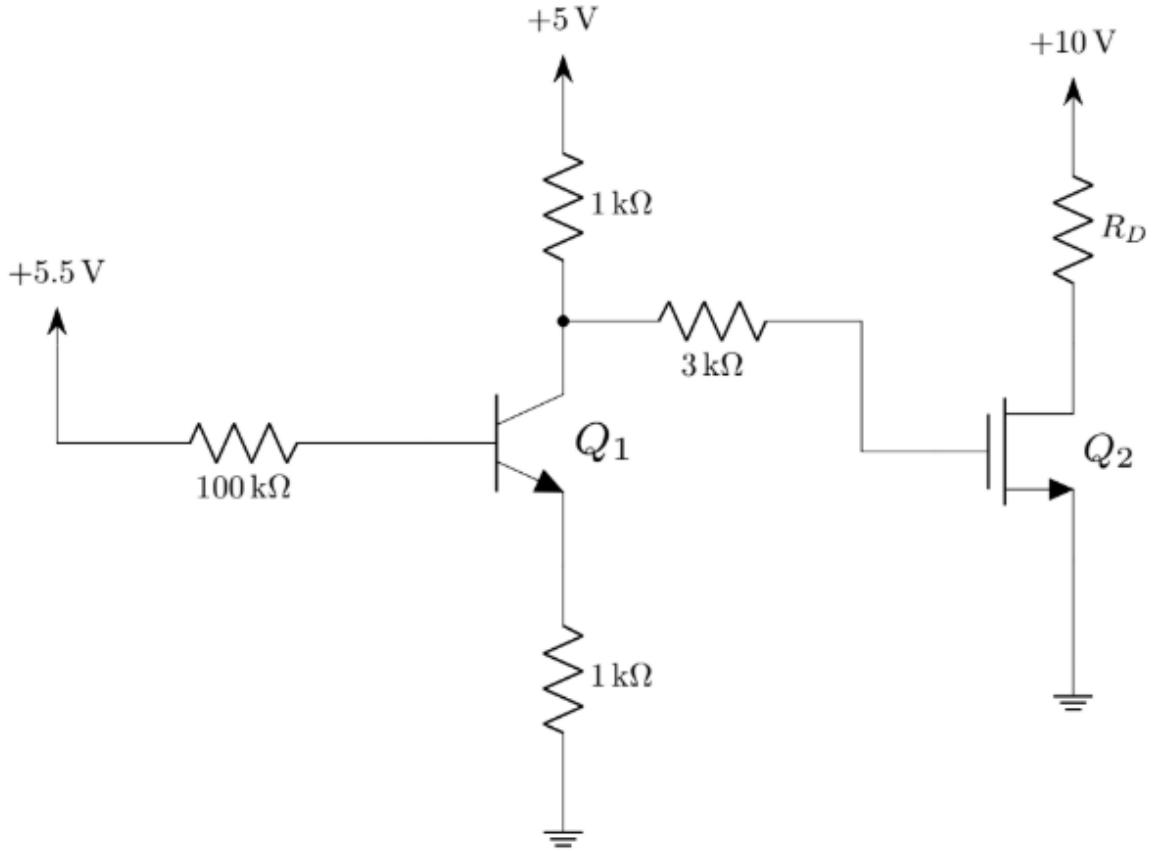
- For the NPN BJT in the adjacent circuit, $\beta = 100$, $V_{BE,\,active} = 0.7\,V$, $V_{BE,\,sat} = 0.8\,V$, and $V_{CE,\,sat} = 0.2\,V$. Determine the operating region of the transistor.



Ans: $\boldsymbol{Sat;\ I_C = 3.9\,mA; I_E = 4\,mA; I_B = 0.1\,mA; V_B = 2.8\,V;\ V_C = 2.2\,V; V_E = 2\,V}$

Problem 12

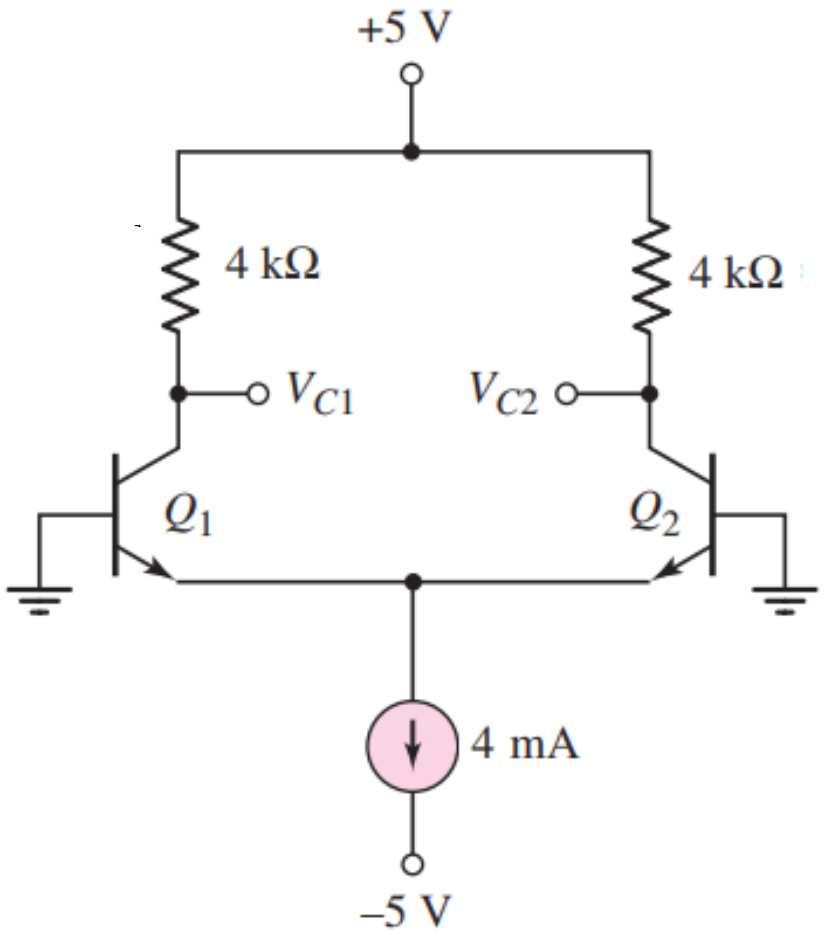
- For the transistor Q_1 in the adjacent circuit, $\beta = 100$, V_{BE} is constant at $0.7\,V$ when Q_1 remains ON, and $V_{CE,\,sat} = 0.2\,V$. For Q_2, $k_n' = 0.25\ {}^{mA}/_{V^2}$, $\left(\frac{W}{L}\right) = \frac{0.72\,\mu m}{0.18\,\mu m}$, and $V_{Tn} = 1\,V$. Determine the value of R_D so that Q_2 operates at the edge of saturation.



Ans: $\boldsymbol{Q_1}$: $\boldsymbol{Active;\ I_C = 2.39\,mA; I_E = 2.4139\,mA; I_B = 23.9\,\mu A; V_B = 3.11\,V;\ V_C = 2.61\,V; V_E = 2.4139\,V}$	Ans: $\boldsymbol{Q_2}$: $\boldsymbol{V_G = 2.61\,V; V_D = 1.61\,V;\ I_{DS} = 1.2961\,mA;\ R_D \approx 7\,k\Omega}$

Problem 13

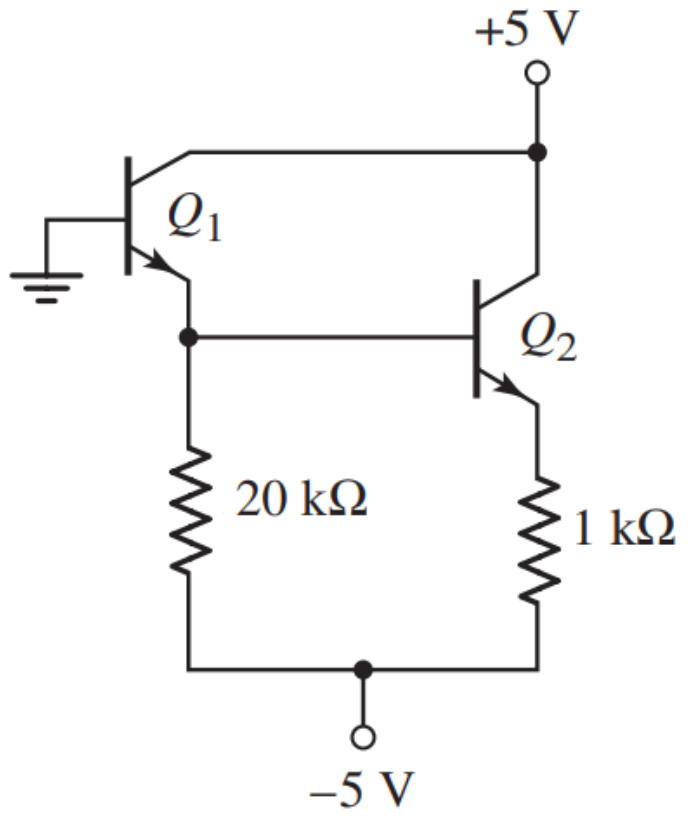
- The transistors in the adjacent circuit are identical. Here, $\beta = 100$, $V_{BE,active} = 0.7\,V$, $V_{BE,sat} = 0.8\,V$, and $V_{CE,sat} = 0.2\,V$. Determine V_{C1} and V_{C2}.



Ans: $\boldsymbol{Both\ Q_1\ \&\ Q_2\ are\ in\ Sat\ Mode;\ I_{E_1} = I_{E_2} = 2\ mA; V_{B_1} = V_{B_2} = 0\ V; V_{E_1} = V_{E_2} = -0.8\ V; V_{C_1} = V_{C_2} = -0.6\ V;\ I_{C_1} = I_{C_2} = 1.4\ mA; I_{B_1} = I_{B_2} = 0.6\ mA}$

Problem 14

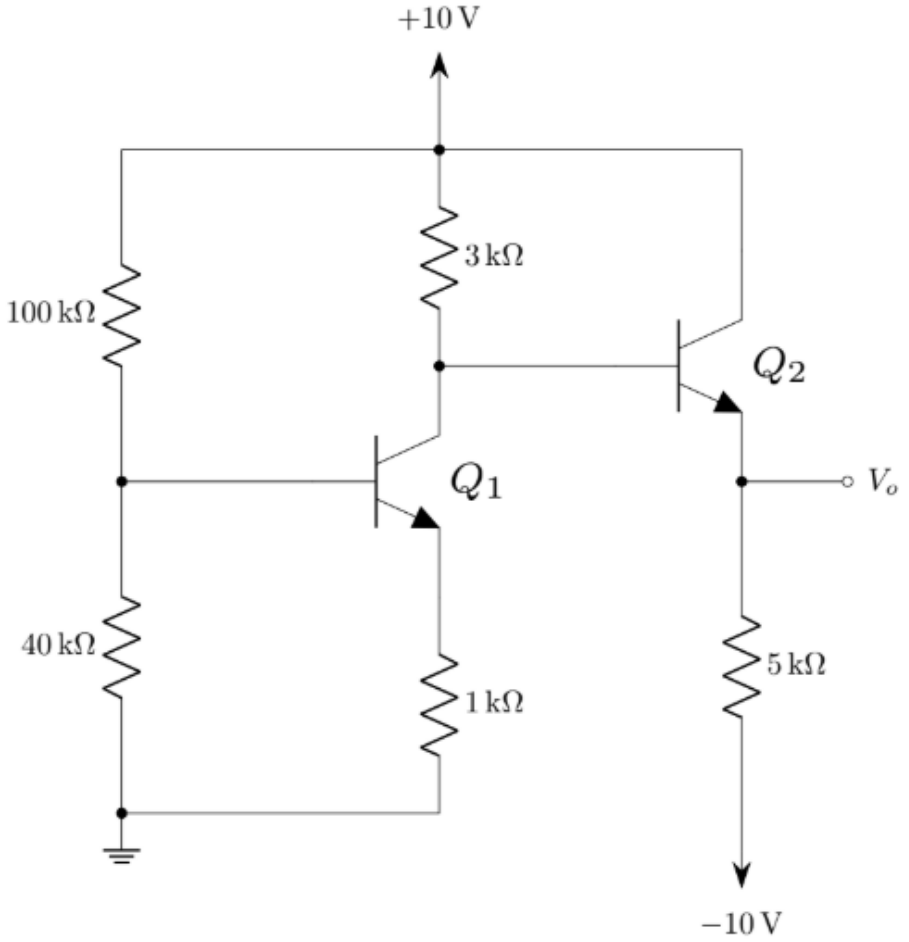
- The transistors in the following circuit are identical. Here, $\beta = 80$, $V_{BE,\,active} = 0.7\,V$, $V_{BE,\,sat} = 0.7\,V$, and $V_{CE,\,sat} = 0.2$. Determine the base currents.



Ans: $\boldsymbol{Both\ Q_1\ \&\ Q_2\ are\ in\ Active\ Mode;\ V_{B_1} = V_{B_2} = 0\,V; V_{E_1} = V_{E_2} = -0.7\,V; I_{E_1} = I_{E_2} = 0.215\,mA; I_{B_1} = I_{B_2} = 2.7\,\mu A;\ V_{C_1} = V_{C_2} = 5\,V;\ I_{C_1} = I_{C_2} = 0.212\,mA}$

Problem 15

- The transistors in the adjacent circuit are identical. Here, $\beta = 120$, $V_{BE,\,active} = 0.7\,V$, $V_{BE,\,sat} = 0.7\,V$, and $V_{CE,\,sat} = 0.2\,V$. Determine V_o.



Ans: $\boldsymbol{Q_1}$ ***is in Active Mode***; $\boldsymbol{I_{B_1} = 0.0238\,mA; I_{C_1} = 2.856\,mA; I_{E_1} = 2.88\,mA; V_{E_1} = 2.88\,V;\ V_{B_1} = 3.58\,V;\ V_{C_1} = V_{B_2} = 1.41\,V;}$
$\boldsymbol{Q_2}$ ***is in Active Mode***: $\boldsymbol{V_{C_2} = 10\,V; I_{B_2} = 0.0177\,mA; I_{C_2} = 2.124\,mA; I_{E_2} = 2.1417\,mA; V_{E_2} = V_0 = 0.71\,V; V_{B_2} = V_{C_1} = 1.41\,V}$

Up Next

Relevant *Questions* from Past Semesters

Final – Fall 2025 - 1/2

- The transistors Q_2 and Q_3 in the following circuit are identical and have the parameters given in the box.
  a. Determine V_1 and V_2.
  b. Determine V_3 and I_{DS}.

NPN BJT Parameters:

For the transistor Q_2,

Common emitter current gain: $\beta = 100$

Base-emitter voltage in the active region: $V_{BE(active)} = 0.7\,\text{V}$

Base-emitter voltage in the saturation region: $V_{BE(sat)} = 0.8\,\text{V}$

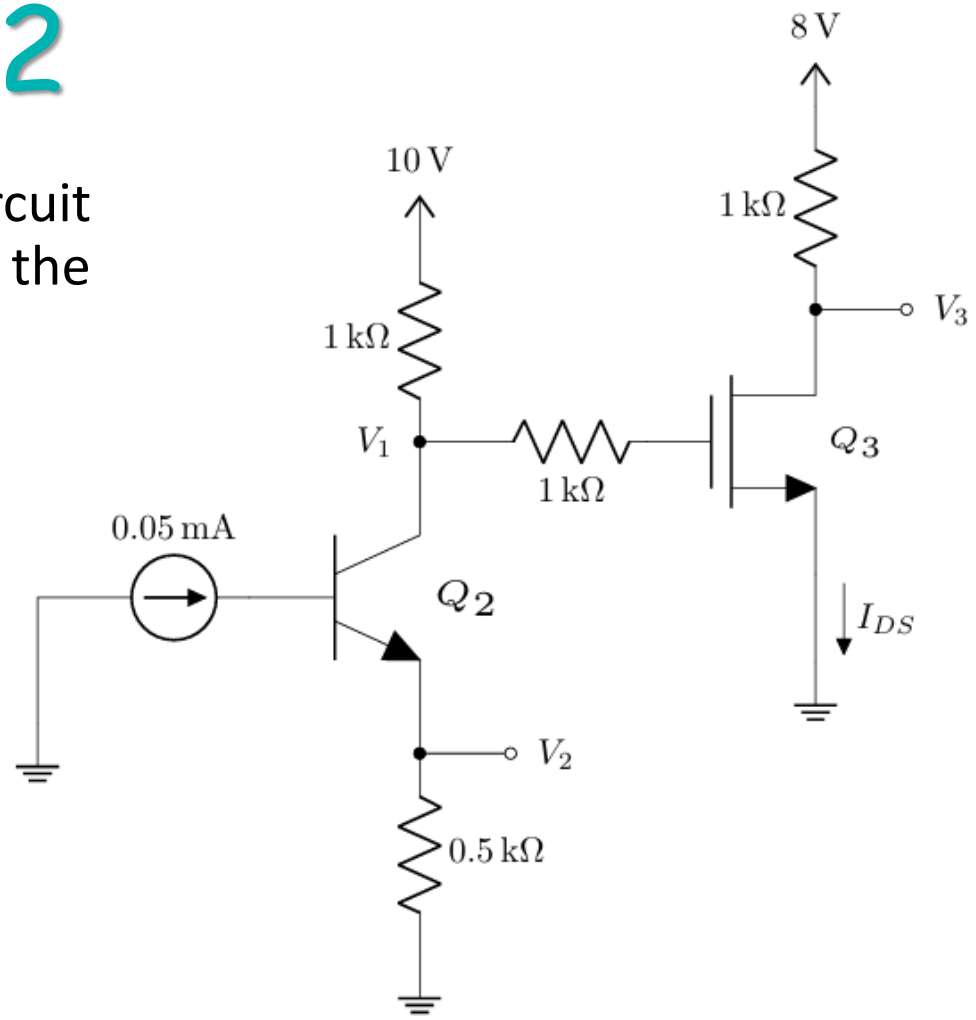
Collector-emitter voltage in the saturation region: $V_{CE(sat)} = 0.2\,\text{V}$

N-Channel MOSFET Parameters:

For the transistor Q_3,

Threshold voltage: $V_{T_n} = 1\,\text{V}$

Transconductance parameter: $k_n = 2\,\text{mA/V}^2$



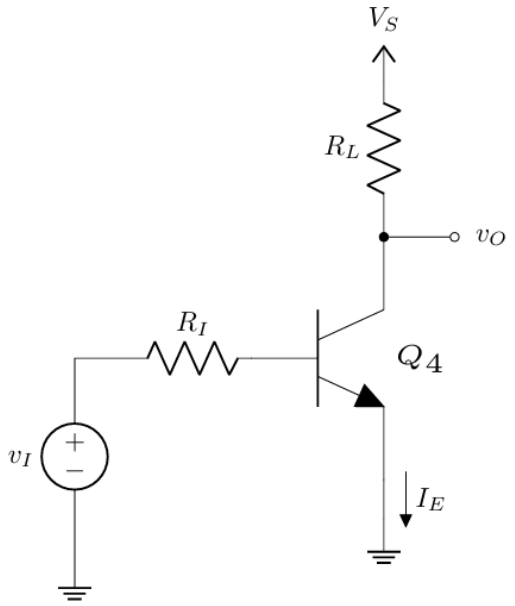
Ans: $(a)\ Q_2 \rightarrow Active;\ V_1 = 5\,V; V_2 = 2.525\,V$
$(b)\ Q_3 \rightarrow Triode; V_3 = 1\,V; I_{DS} = 7\,mA$

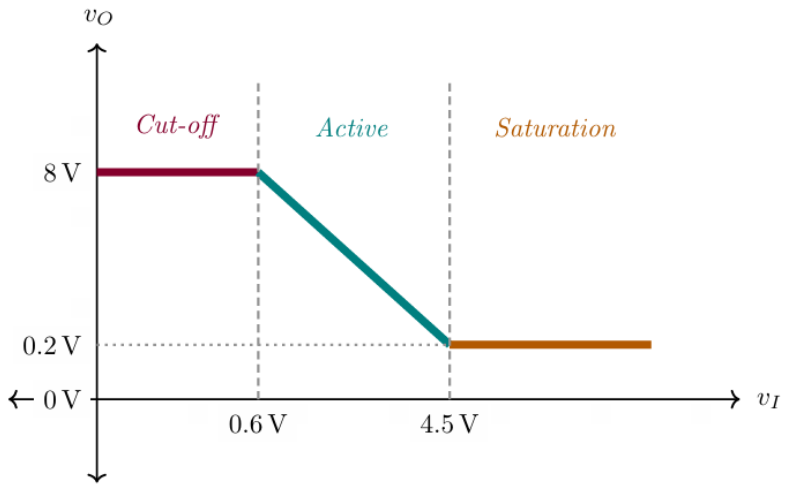
Final – Fall 2025 - 2/2

The output voltage v_O for the circuit shown below is plotted as a piecewise function of the input voltage v_I. The different regions of transistor operation are indicated by vertical dashed lines, and the specific transistor parameters are provided in the accompanying box.

a. When the transistor operates at the edge of saturation, the emitter current is $I_E = 3.94\,mA$. Determine the values of R_L and R_I. [Use $\beta = 100$]

b. Using the values found in part (a), what is the value of I_E when $v_I = 5\,V$.





NPN BJT Parameters:

For the transistor Q_4,

Common emitter current gain: $\beta = 100$

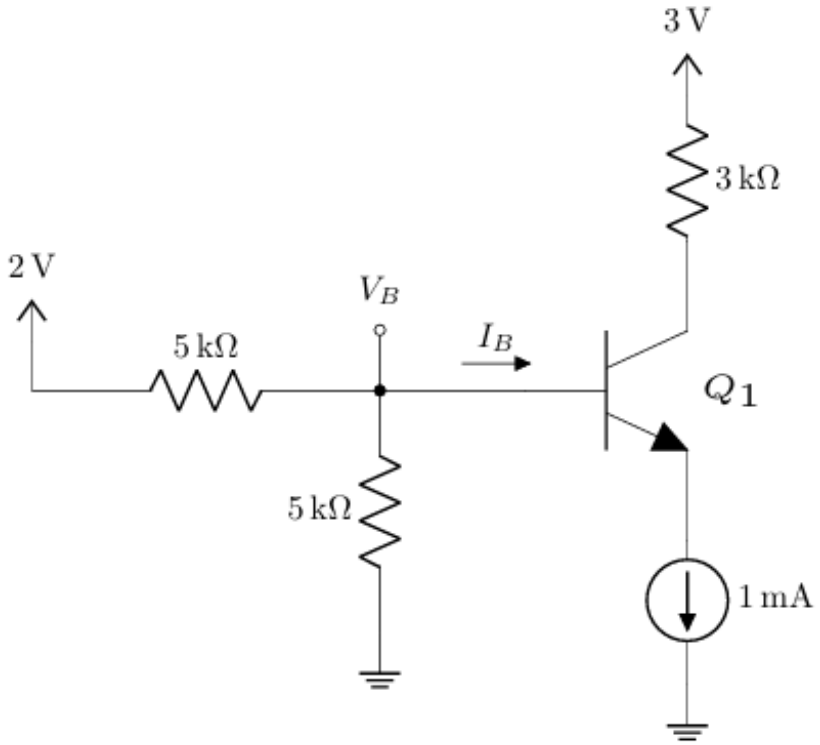
Base-emitter voltage: $V_{BE(sat)} = V_{BE(active)} \approx 0.6\,\text{V}$

Collector-emitter voltage: $V_{CE(sat)} = V_{CE(edge\ of\ sat)} \approx 0.2\,\text{V}$

Ans: (a) $\boldsymbol{R_I = 100\ k\Omega; R_L = 2\ k\Omega}$
(b) $\boldsymbol{I_E = 3.944\ mA; I_B = 0.044\ mA; I_C = 3.9\ mA}$

Final – Summer 2025

- Verify that Q_1 operates in the saturation region, and determine all terminal currents and voltages of the transistor.



NPN BJT Parameters:

For the transistor Q_1:

Current gain: $\beta = 100$

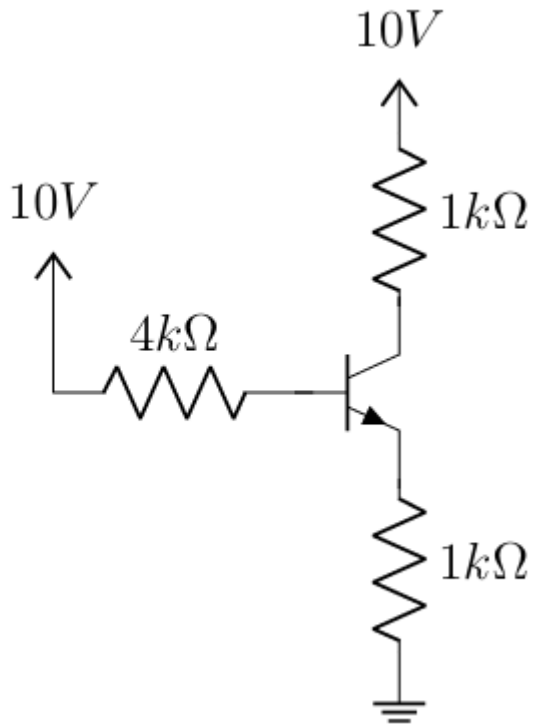
Base-emitter voltage in saturation: $V_{BE(sat)} = 0.8\,\text{V}$

Collector-emitter voltage in saturation: $V_{CE(sat)} = 0.2\,\text{V}$

Ans: $\boldsymbol{Sat;\ I_C = 0.927\,mA; I_E = 1\,mA; I_B = 0.073\,mA; V_B = 0.818\,V;\ V_C = 0.218\,V; V_E = 0.018\,V}$

Final – Spring 2025

- Analyze the circuit below, and calculate I_B, I_C, I_E, V_C, and V_E using the method of assumed states. You must validate your assumption.



NPN BJT Parameters:

For the transistor Q_1:

Current gain: $\beta = 100$

Base-emitter voltage in saturation: $V_{BE(sat)} = 0.8\,\text{V}$

Collector-emitter voltage in saturation: $V_{CE(sat)} = 0.2\,\text{V}$

Ans: $\boldsymbol{Sat;\ I_C = 4.42\,mA; I_E = 5.38\,mA; I_B = 0.96\,mA; V_B = 6.18\,V;\ V_C = 5.58\,V; V_E = 5.38\,V}$

Final – Fall 2024

- Analyze the circuit below, and calculate I_B, I_C, I_E, and V_O using the method of assumed states. You must validate your assumption.

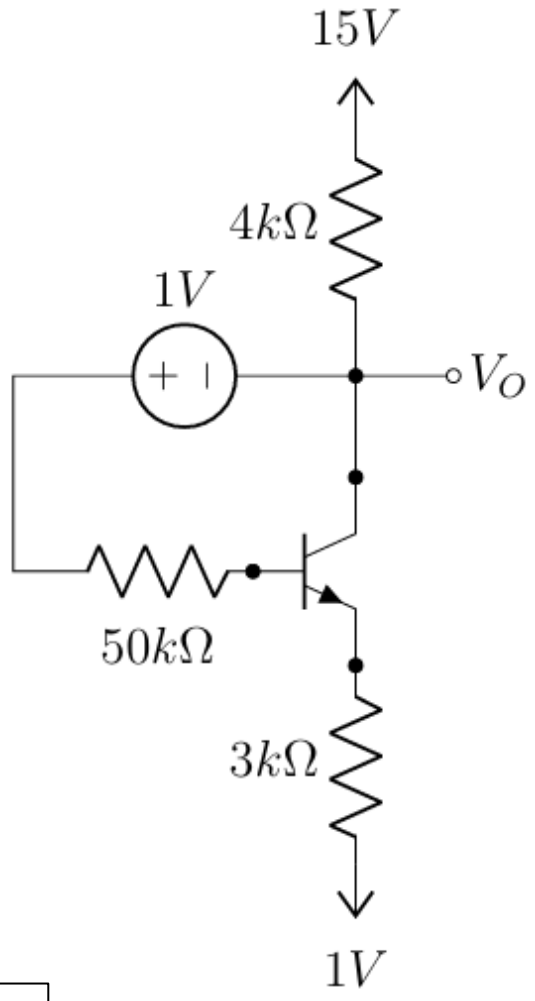
NPN BJT Parameters:

For the transistor Q_1:

Current gain: $\beta = 100$

Base-emitter voltage in saturation: $V_{BE(sat)} = 0.8\,\text{V}$

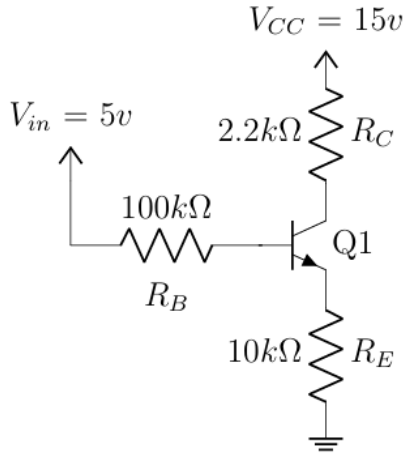
Collector-emitter voltage in saturation: $V_{CE(sat)} = 0.2\,\text{V}$



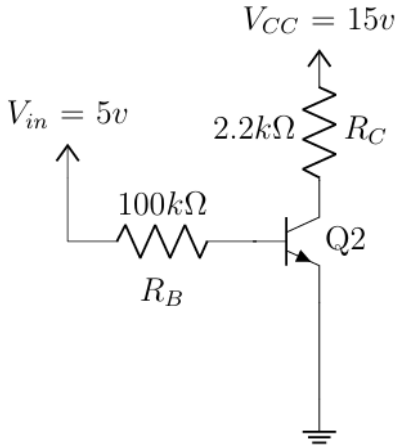
Ans: $\boldsymbol{Active;\ I_C = 1.89\,mA; I_E = 1.908\,mA; I_B = 0.0189\,mA; V_B = 5.515\,V;\ V_C = 5.46\,V; V_E = 4.715\,V}$

Final – Summer 2024 - 1/2

- Karim designed a BJT circuit. When his friend Rahim saw it, he claimed that the BJT would operate in the Active mode. Then Rahim built a circuit and asked Karim to figure out the operating mode the BJT.
  a. Calculate I_B, I_C, I_E, and V_{CE} in Karim's circuit and determine whether Rahim was correct or not.
  b. Analyze Rahim's circuit and calculate I_B, I_C, I_E, and V_{CE} to help Karim determine the operating state of the BJT.



Karim's Circuit



Rahim's Circuit

For Both BJTs

$\beta = 100$

$V_{BE(Active)} = 0.7v$

$V_{BE(Saturation)} = 0.8v$

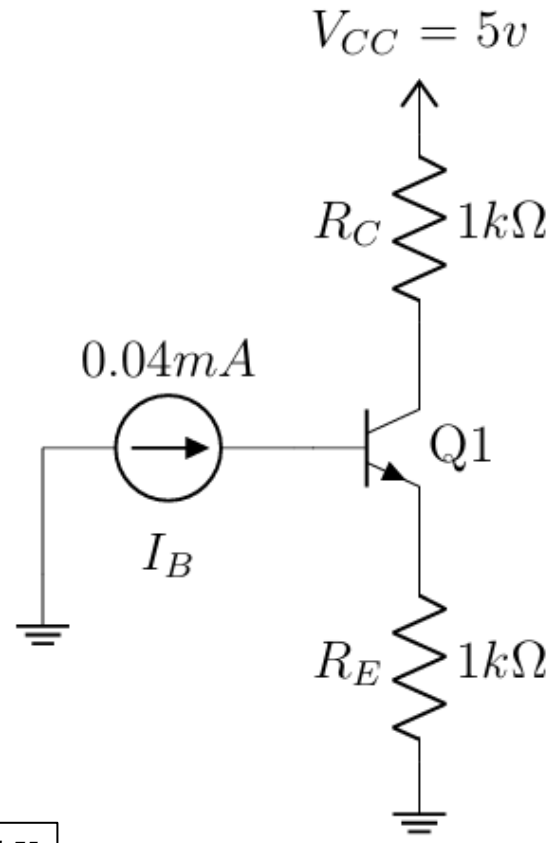
$V_{CE(Saturation)} = 0.2v$

Ans: (a) ***Active***; $I_C = 0.3874\ mA; I_E = 0.391\ mA; I_B = 3.874\ \mu A; V_B = 4.61\ V;\ V_C = 14.15\ V; V_E = 3.91\ V$
(b) ***Active***; $I_C = 4.3\ mA; I_E = 4.343\ mA; I_B = 0.043\ mA; V_B = 4.61\ V;\ V_C = 5.54\ V; V_E = 0\ V$

Final – Summer 2024 - 2/2

- Analyze the circuit below, and calculate I_B, I_C, I_E, and V_{CE} using the method of assumed states. You must validate your assumption.

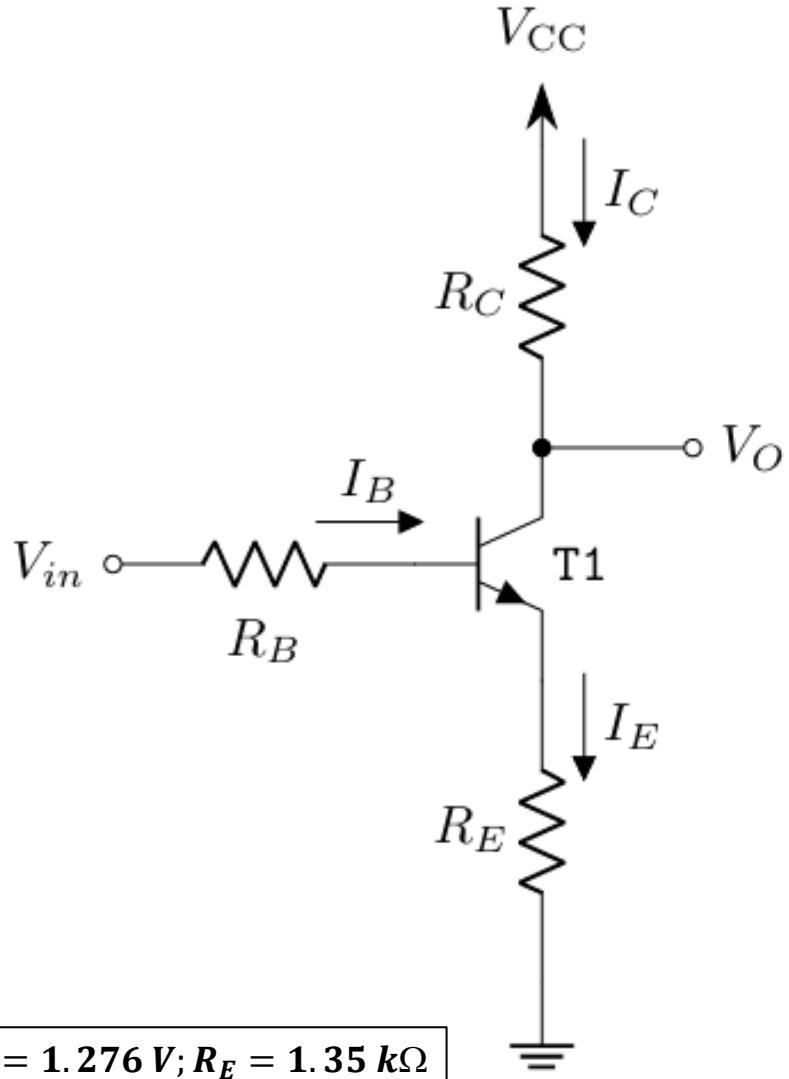
$$\underline{\text{BJT Parameters}}$$
$$\beta = 90$$
$$V_{BE(Active)} = 0.7v$$
$$V_{BE(Saturation)} = 0.8v$$
$$V_{CE(Saturation)} = 0.2v$$



Ans: $\boldsymbol{Sat;\ I_C = 4.93\ mA; I_E = 4.9825\ mA; I_B = 0.0525\ mA; V_B = 5.7825\ V;\ V_C = 5.46\ V; V_E = 4.715\ V}$

Final – Spring 2024 - 1/2

- Andrew and Nicole found the adjacent circuit built in a trainer board. From the transistor model, they knew it had a gain of $\beta = 80$. Additionally, $V_{BE,\,active} = 0.7\,V$, $V_{BE,\,sat} = 0.7\,V$, and $V_{CE,\,sat} = 0.2\,V$. They also saw $V_{CC} = 5\,V$, $R_B = 2\,k\Omega$, and $R_C = 3\,k\Omega$. However, the R_E resistor was an unknown one. So, they provided an input of $V_{in} = 2\,V$ and measured the output to be $V_O = 2.2\,V$. Nicole said, "In this condition, the transistor is in active mode". But, Andrew disagreed.
  a. Design the circuit, i.e., determine the value of R_E, using what Nicole said about the mode of the transistor.
  b. Using the value of R_E obtained from (a), determine who will be right if $V_{in} = 4\,V$.



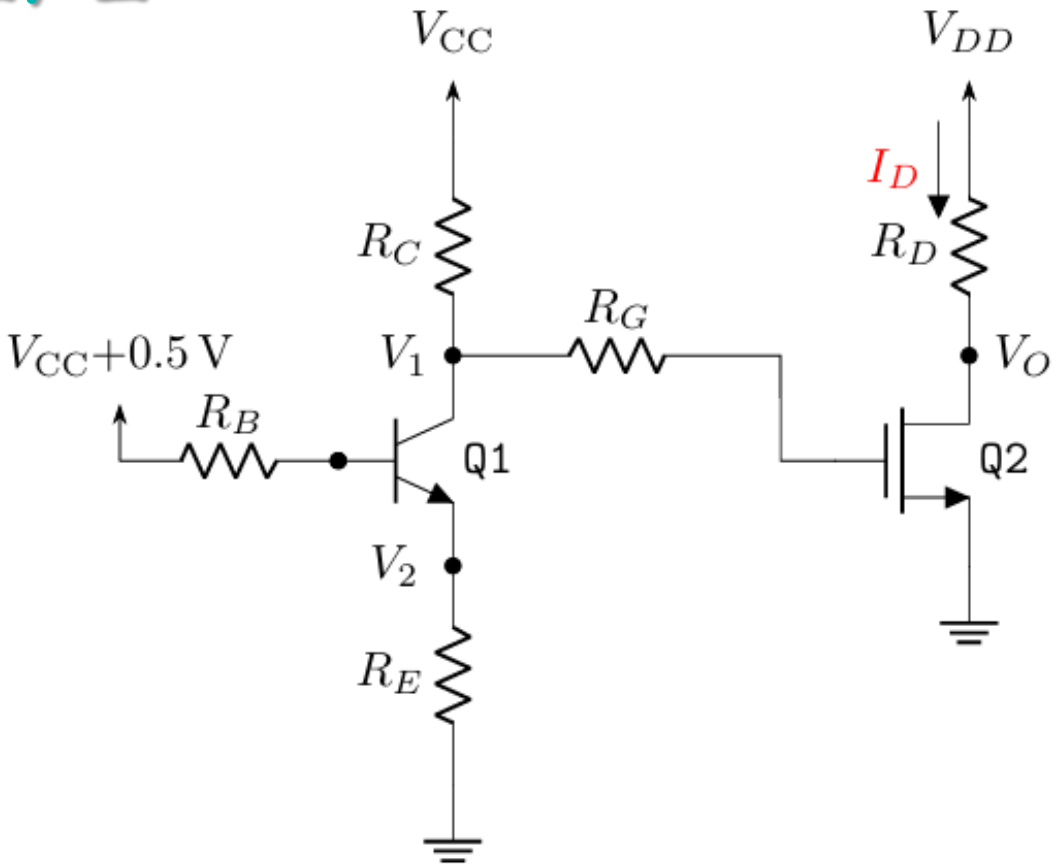
Ans: (a) $I_C = 0.933\,mA; I_E = 0.945\,mA; I_B = 0.012\,mA; V_B = 1.976\,V;\ V_C = V_O = 2.2\,V; V_E = 1.276\,V; R_E = 1.35\,k\Omega$
(b) $Sat;\ I_C = 0.912\,mA; I_E = 1.53\,mA; I_B = 0.618\,mA; V_B = 2.766\,V;\ V_C = V_O = 2.264\,V; V_E = 2.066\,V;$

Final – Spring 2024 - 2/2

- For the adjacent circuit,
  a. Find out the currents going through the three terminals of the transistor Q1.
  b. Determine the the value of V_1 and V_2.
  c. Design the circuit, i.e. find an appropriate value of R_D, such that Q2 remains at the edge of saturation, i.e., when $V_{DS} = V_{GS} - V_T$.

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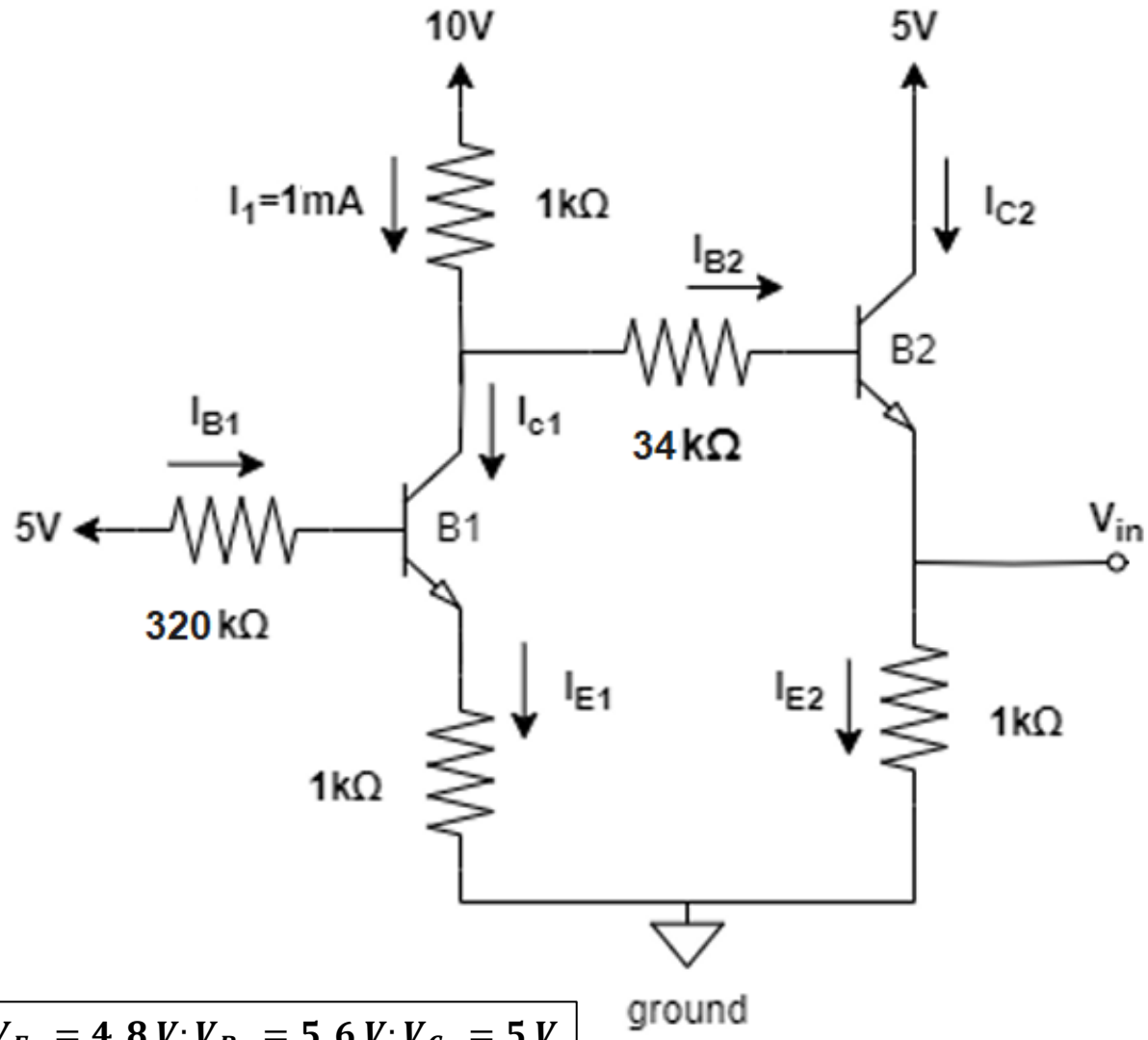


Resistor values in (kΩ)					Supply Voltages
R_B	R_C	R_E	R_G	R_D	V_{CC} =5 V
100	1.1	1	3	-	V_{DD} =10 V

Ans: (a) Sat; $I_C = 2.27\ mA; I_E = 2.295\ mA; I_B = 0.025\ mA; V_B = 3\ V$
(b) $V_1 = V_C = 2.503\ V; V_2 = V_E = 2.295\ V$
(c) $V_D = 1.5\ V; R_D = 7.556\ k\Omega$

Final – Fall 2023

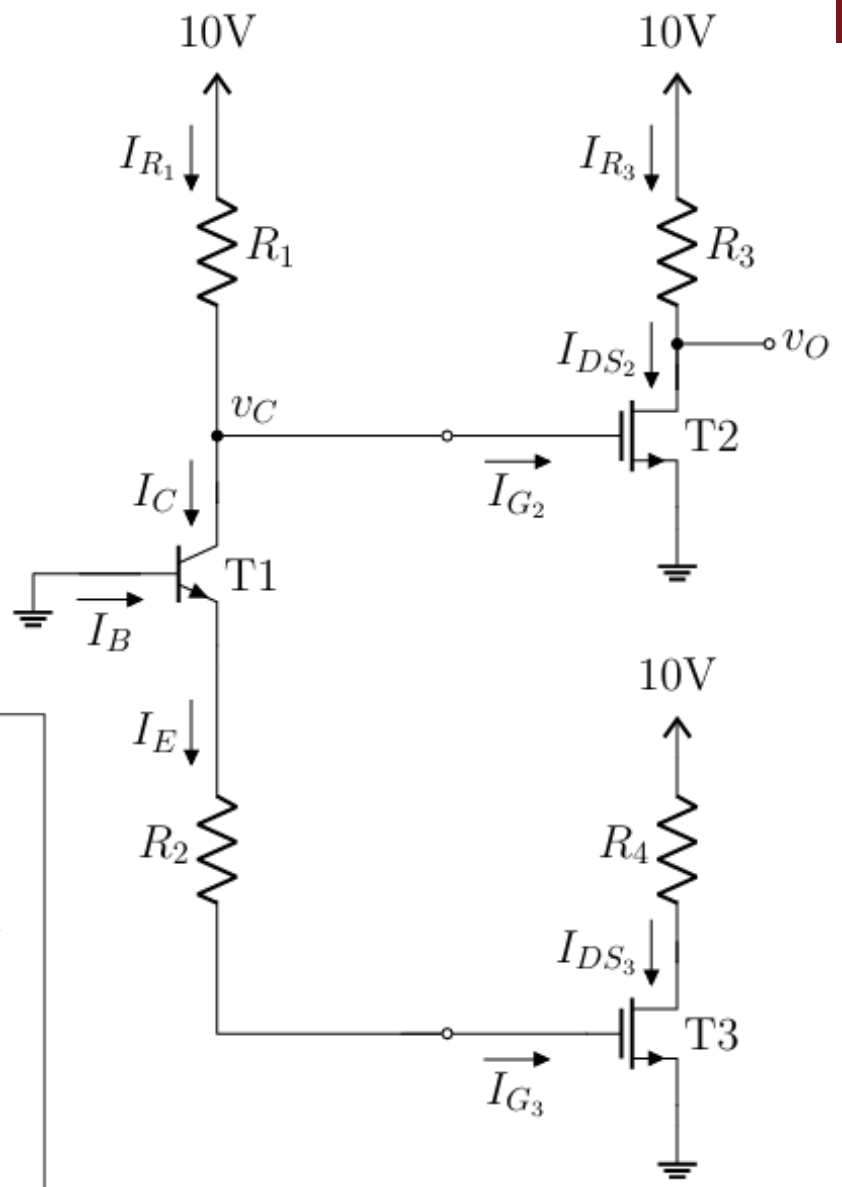
- The BJTs in the following circuit have parameters: $\beta = 75$, $V_{BE,active} = V_{BE,sat} = 0.8\,V$, and $V_{CE,sat} = 0.2\,V$. Determine the operating regions of B_1 and B_2.



Ans: $\boldsymbol{B2 \rightarrow Sat;\ I_{E_2} = 4.8\,mA; I_{C_2} = 4.7\,mA; I_{B_2} = 0.1\,mA; V_{E_2} = 4.8\,V; V_{B_2} = 5.6\,V; V_{C_2} = 5\,V}$
$\boldsymbol{B1 \rightarrow Active;\ I_{E_1} = 0.912\,mA; I_{C_1} = 0.9\,mA; I_{B_1} = 0.012\,mA; V_{E_1} = 0.912\,V; V_{B_1} = 1.16\,V; V_{C_1} = 9\,V}$

Final – Summer 2023

- For the adjacent circuit, $R_1 = 1\ k\Omega$, $R_2 = 2\ k\Omega$, $R_3 = 3\ k\Omega$, and $R_4 = 4\ k\Omega$.
  a. Determine the values of I_C, I_B, I_E, and v_C of T1.
  b. Determine the value of I_{DS_2} and v_O of T2.



for BJT
$\beta = 100$
$\alpha = 0.99$
$v_{BE(Active)} = 0.7V$
$v_{BE(Saturation)} = 0.8V$
$v_{CE(Saturation)} = 0.2V$

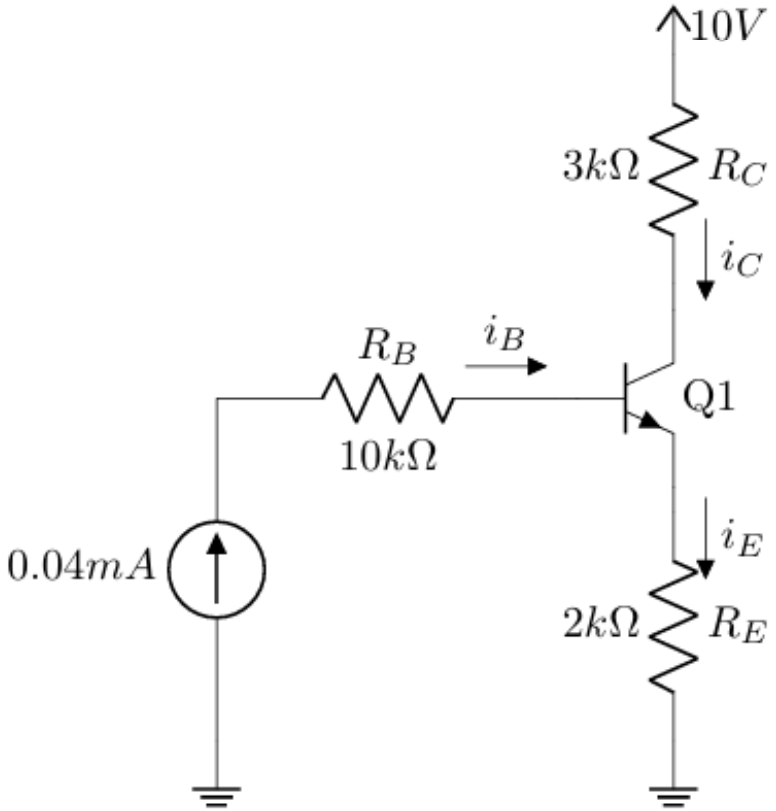
for MOSFET
$V_T = 1V$
$k = 5mA/V^2$

Ans: **T1 → *Cutoff*; $I_E = I_B = I_C = 0\ mA$; $v_C = 10\ V$; $V_E > 0.8\ V$; $V_B = 0\ V$;**
T3 → *Curoff*; $I_{DS_2} = 0\ mA$; $I_{G_3} = 0\ mA$
T2 → *Triode*; $I_{G_2} = 0\ mA$; $I_{DS_2} = 3.31\ mA$; $v_O = V_D = 0.074\ V$

Final – Spring 2023 - 1/3

- Analyze the following circuit to find the values of i_B, i_C, i_E, and v_{CE}. Assume saturation mode and validate the assumptions.

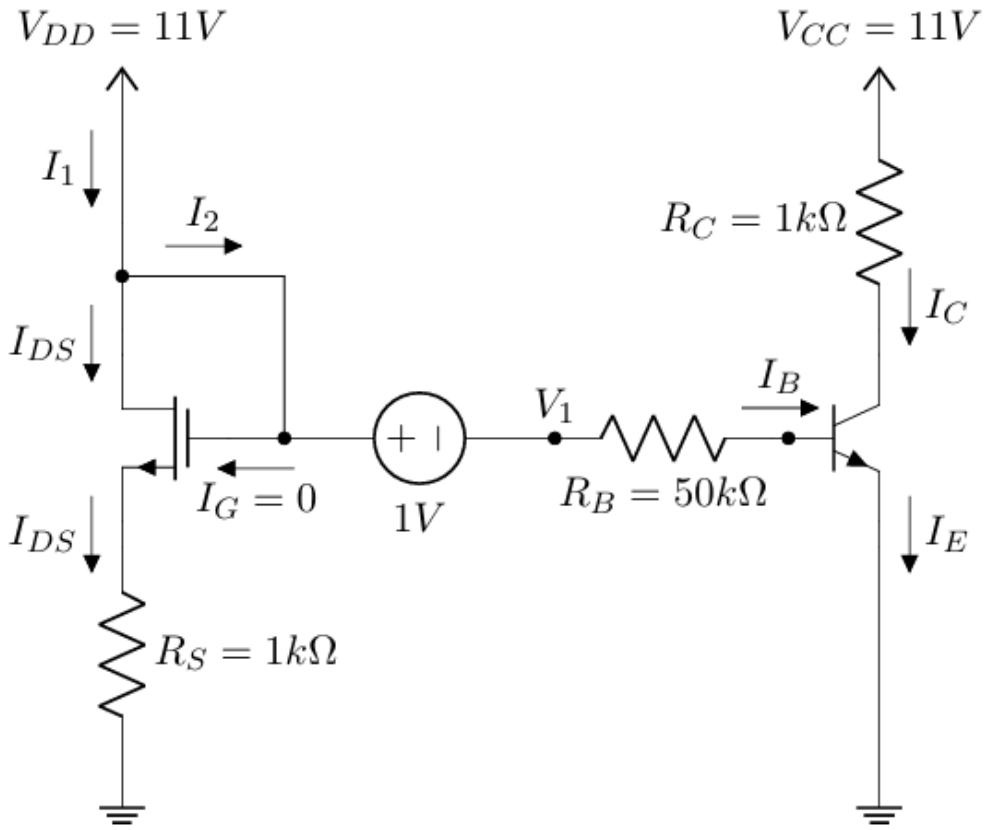
$$\beta = 100$$
$$\alpha = 0.99$$
$$v_{BE(Active)} = 0.7V$$
$$v_{BE(Saturation)} = 0.8V$$
$$v_{CE(Saturation)} = 0.2V$$



Ans: $\boldsymbol{Sat;\ i_C = 1.944\ mA; i_E = 1.984\ mA; i_B = 0.04\ mA; V_B = 4.768\ V;\ V_C = 4.168\ V; V_E = 3.968\ V}$

Final – Spring 2023 - 2/3

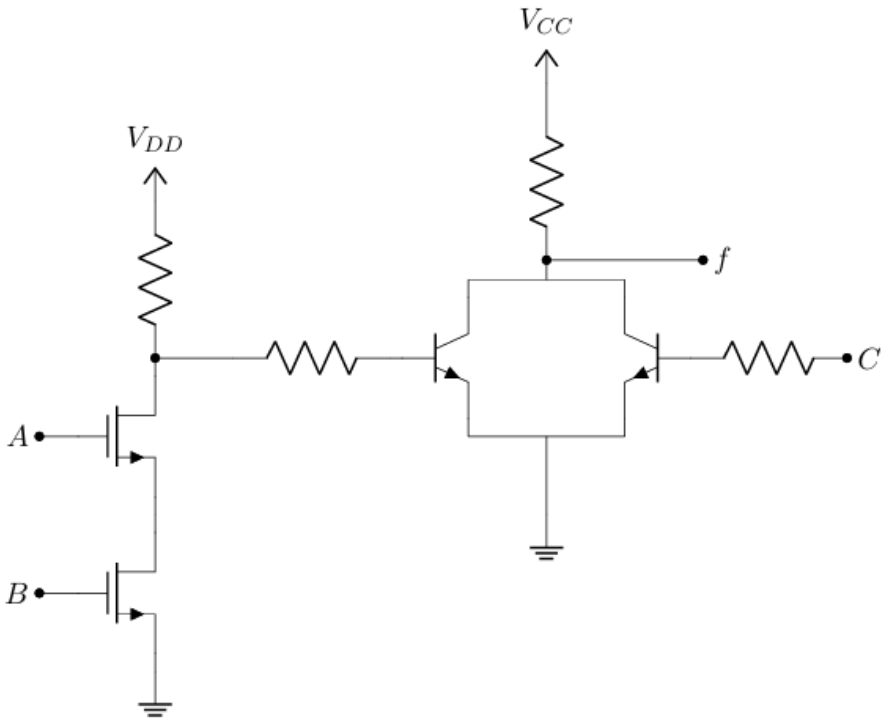
- The transistors in the following circuit have parameters: $k_n = 5\ {}^{mA}/_{V^2}$, $V_{T_n} = 0.7\ V$, $\beta = 85$, $V_{BE,\,active} = 0.7\ V$, $V_{BE,\,sat} = 0.8\ V$, and $V_{CE,\,sat} = 0.2\ V$.
  a. Find out the operating mode of the MOSFET. *[Hint: You don't need to solve the circuit to proof that.]*
  b. Calculate I_{DS} and V_{DS} using the given parameters.
  c. Determine V_1, I_B, I_C, and I_E.



Ans: (b) $\boldsymbol{Sat;\ I_{DS} = 8.46\ mA; V_{DS} = 8.46\ V}$
$\boldsymbol{(c)\ Sat;\ V_1 = 10\ V; I_B = 0.184\ mA; I_C = 10.8\ mA; I_E = 10.984\ mA;\ V_C = 0.2\ V; V_B = 0.8\ V; V_E = 0\ V}$

Final – Spring 2023 - 3/3

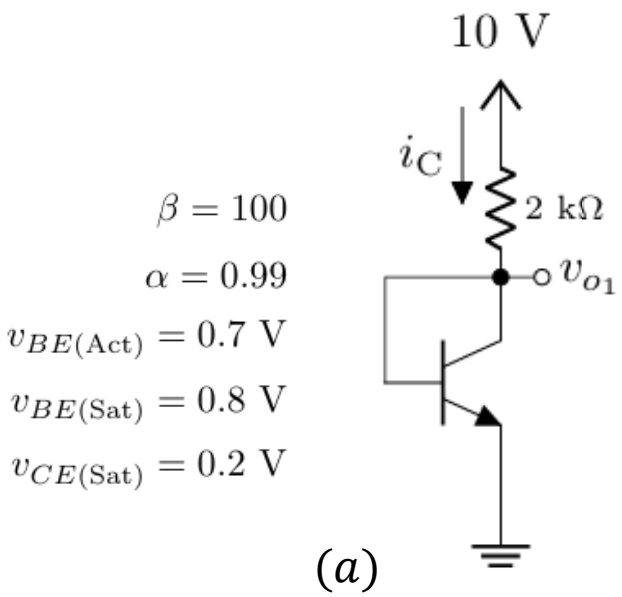
a. For the logic circuit (b), write an expression for the Boolean output f in terms of the Boolean inputs A, B, and C.



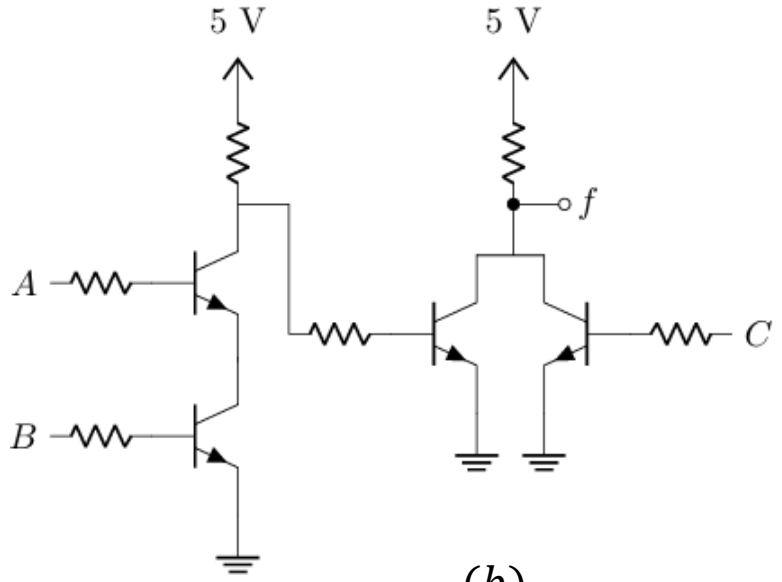
Final – Fall 2022

a. Analyze the circuit (a) to find i_C and v_{O1} using the Method of Assumed State. Validate your assumptions.

b. For the logic circuit (b), write an expression for the Boolean output f in terms of the Boolean inputs A, B, and C.



(a)

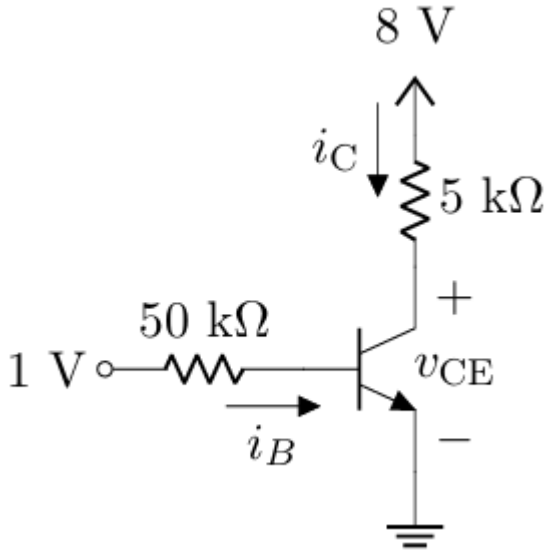


(b)

Ans: (a) $\boldsymbol{Active;\ i_C = 4.65\ mA; i_E = 4.6965\ mA; I_B = 0.0465\ mA; V_B = V_C = v_{o1} = 0.7\ V;\ V_E = 0\ V}$

Final – Summer 2022

- Analyze the adjacent circuit to find i_C and v_{CE} using the Method of Assumed State. Validate your assumptions.



$\beta = 100$

$\alpha = 0.99$

$v_{BE(\text{Active})} = 0.7\ \text{V}$

$v_{BE(\text{Saturation})} = 0.8\ \text{V}$

$v_{CE(\text{Saturation})} = 0.2\ \text{V}$

Ans: $\boldsymbol{Active;\ i_C = 0.6\ mA; i_E = 0.6006\ mA; I_B = 0.006\ mA; V_B = 0.7\ V;\ V_E = 0\ V; V_C = V_{CE} = 5\ V}$

Acknowledgement and References

Some of the problems in this set are taken or adapted from the following sources:

1. Agarwal, A. and Lang, J., Foundations of Analog and Digital Electronic Circuits, Morgan Kaufmann
2. Sedra, A. S., & Smith, K. C., Microelectronic Circuits, Oxford University Press
3. Neamen, D. A., Microelectronics: Circuit Analysis and Design, McGraw-Hill

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